Predicting Diabetes Status Using Health Factors

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Github*:* https://github.com/vaishuc17/data1030finalproject.git

1 Introduction

Diabetes is a chronic condition where blood sugar levels remain high due to the body not producing enough insulin or not utilizing it effectively and, if left untreated, can lead to the development of other health problems, such as heart disease [1]. In 2021, 38.1 million U.S. adults (18 years or older) had diabetes and an additional 8.7 million adults who met the criteria for diabetes were either unaware or did not report having diabetes [2]. These statistics highlight the prevalence of diabetes in society and also motivate a need for machine learning tools that could make diagnoses easier and more accurate.

Chowdhury et al. used the 2021 Behavioral Risk Factor Surveillance System (BRFSS) dataset to predict diabetes vs. no diabetes [3]. Due to the dataset being imbalanced (more cases of people without diabetes than with diabetes), they tried various sampling techniques, including SMOTE-N (Synthetic Minority Over-sampling Technique) to increase the number of samples in the minority class and ENN (Edited Nearest Neighbors) to reduce the number of samples in the majority class. For Logistic Regression, they achieved an accuracy of 0.62 with ENN, 0.71 with SMOTE-N, and 0.83 with no balancing technique. For Random Forest, they achieved an accuracy of 0.69 with ENN, 0.78 with SMOTE-N, and 0.82 with no balancing technique. For gradient boosting, they achieved an accuracy of 0.70 with ENN, 0.78 with SMOTE-N, and 0.83 with no balancing technique. Although accuracy was higher without any balancing strategy, for all ML models they found increases in recall, with Logistic Regression's recall score increasing by 12.2% with oversampling.

For this project, a cleaned version of the CDC's 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset was used [4]. The BRFSS collects data about U.S. residents' health-related factors and risk behaviors, as well as chronic health conditions, through health-related telephone surveys. The original dataset contains responses from 441,455 individuals and has 330 features, whereas the cleaned version contains 253,680 responses and 21 features, as well as 1 target variable that has 3 classes: 0 (no diabetes), 1 (prediabetes), 2 (diabetes). Thus, this is a multiclass classification problem. The features in the cleaned dataset include health indicators such as BMI and HighBP, as well as demographic factors such as Age and Sex, that can help in predicting diabetes. The goal of this project is to determine whether these factors can be used by ML models to predict whether an individual has diabetes, prediabetes, or no diabetes.

2 Exploratory Data Analysis

Figures showing the distribution of each of the features and the target variable were created to better understand their distribution. Figure 1 below reveals the extreme class imbalance present in the dataset that will need to be taken into account when splitting data.

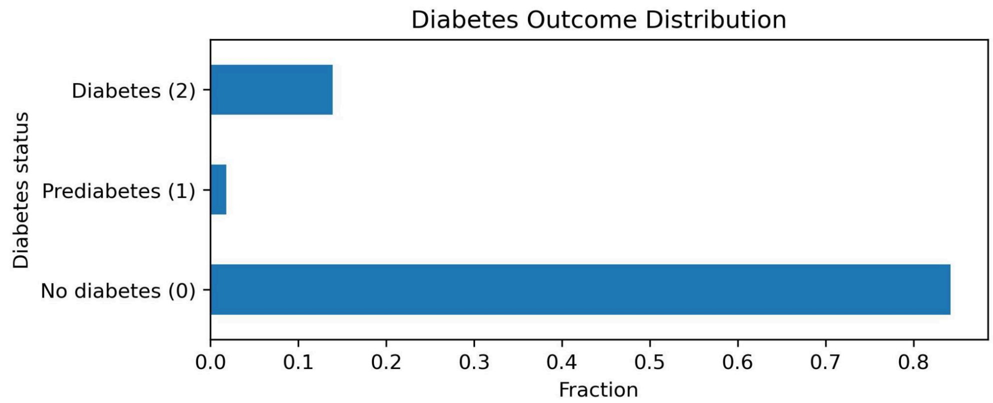


Figure 1. Proportion of each target variable class (approx. 0.14 in class 0, 0.02 in class 1, and 0.84 in class 0)

The correlations between features was also investigated to rule out multicollinearity in Figure 2. As we can see, none of the features are highly correlated with each other.

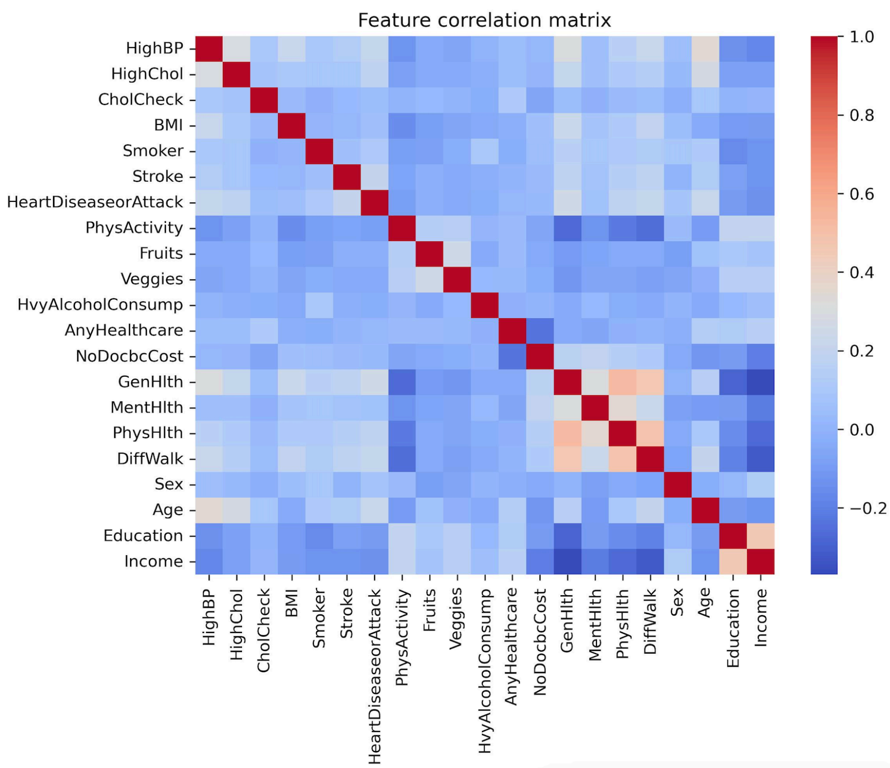


Figure 2. Feature correlation matrix

Since studies have found obesity and old age to be a risk factor for diabetes [5], the distribution of the BMI feature by class and Age feature by class were calculated and are shown in Figure 3 and Figure 4, respectively. In Figure 3, we see that the median BMI for individuals without diabetes is lower than for prediabetes and diabetes. The median for prediabetes and diabetes is relatively similar and their boxes also have high overlap. In Figure 4, we see that as the age group increases, the proportion of individuals in the group that have diabetes increases.

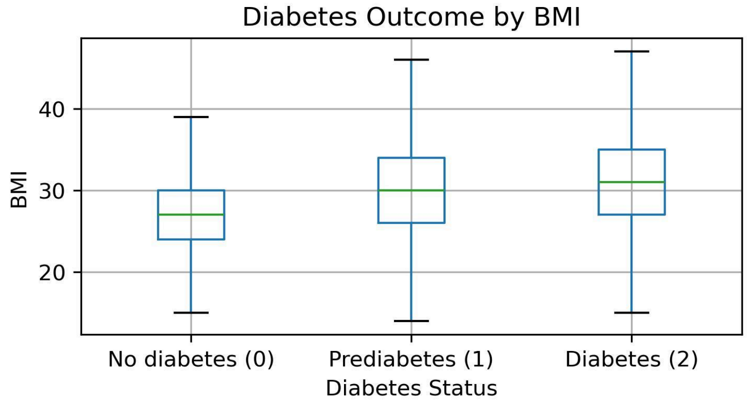


Figure 3. Box plot of the BMI distribution by class

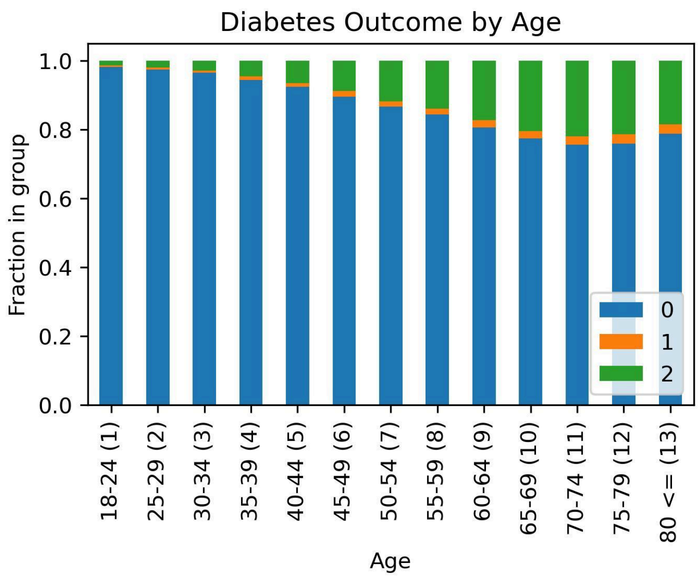


Figure 4. Bar plot depicting the fraction of each group belonging to each of the classes

3 Methods

Two different pipelines were used. Four models, Logistic Regression, Random Forest Classifier, K Neighbors Classifier, and Linear Support Vector Classifier (LinearSVC), were implemented using GridSearchCV, while XGBClassifier was implemented by manually looping through hyperparameters and folds in order to incorporate early stopping. Due to the dataset having an imbalanced class distribution, stratification was used in the splitting strategy.

For both pipelines, with and without GridSearchCV, train_test_split, with stratification on the target variable, was used to split 5% of the data into a test set and the remaining 95% was used for the train and validation set. Since there are over 253,680 observations in the dataset, 5% still leaves 12,684 observations for us to adequately reason over in the test set. For cross validation, StratifiedKFold was used, with the number of splits set to 3. Thus, approximately 63.33% of the dataset was used for training and 31.67% for validation.

Due to the large size of the dataset, both pipeline functions took a parameter that specified the percentage of the dataset to use, allowing us to test its functionality on a smaller portion of the dataset before proceeding to train, validate, and test on the whole dataset. The values reported below, however, are from using the entire dataset. For data preprocessing, only sklearn's StandardScaler was required since all the data was numerical and no missing values were present. Thus, the number of features remained unchanged.

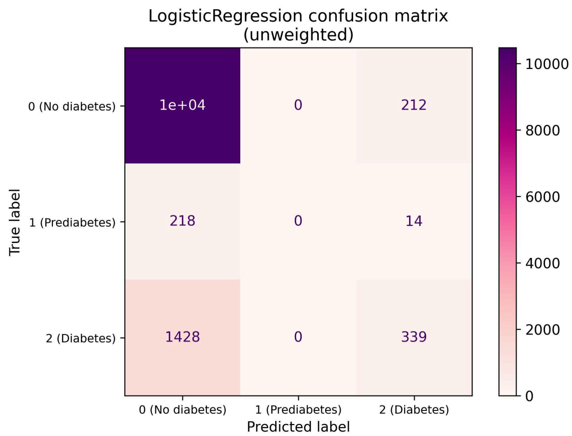
To account for uncertainties in the evaluation metric due to splitting and due to non-deterministic ML methods, both pipelines train, validate, and test over three random states, allowing for replication. Specifically, for each random state, following cross validation, having found the best hyperparameters, we use this best model to make predictions and compute test scores and then compute the mean score across all states. F1 macro was chosen as the evaluation metric since taking the unweighted average of all the classes f1 score is ideal since we have a highly imbalanced dataset and accuracy would not provide a true picture of performance. However, accuracy as well as f1 micro and f1 weighted were also calculated to get a complete picture. Table 1 displays the different ML models used and the parameters tuned, as well as the best parameters for the last random state.

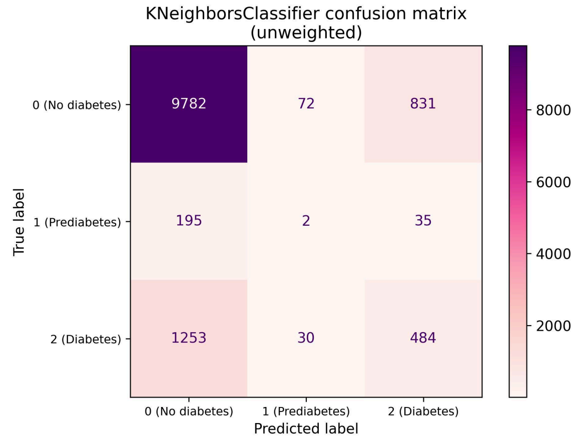
ML Model	**Hyperparameter**	**Best values** *(U=unweighted, W=weighted)*
Logistic Regression	penalty: l1, l2 C: 0.001, 0.01, 0.1, 1.0, 10.0, 100.0, 1000.0	penalty: l1 C: 1.0
Random Forest Classifier	max_depth: 1, 3, 5, 10, 30 max_features: 0.1, 0.3, 0.5, 0.7	max_depth: 30 (U), 10 (W) max_features: 0.7 (U), 0.1 (W)
K Neighbors Classifier	n_neighbors: 3, 5, 7, 9, 13 p: 1, 2 weights: uniform, distance	n_neighbors: 3 p: 1 weights: distance
Linear Support Vector Classifier	C: 0.01, 0.1, 1, 10	C: 0.1 (U), 0.1 (W)
XGB Classifier	learning_rate: 0.05, 0.1 max_depth: 1, 3, 5, 10, 30 colsample_bytree: 0.3, 0.9	learning_rate: 0.1 (U), 0.05 (W) max_depth: 30 (U), 30 (W) colsample_bytree: 0.9 (U), 0.3 (W)

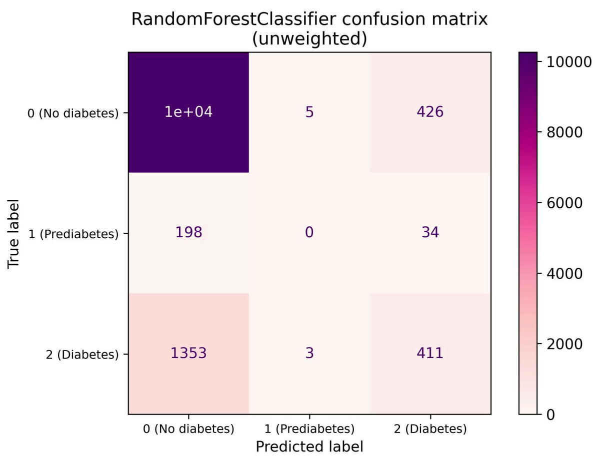
Table 1. Hyperparameter values as well as their best values for each of the ML models

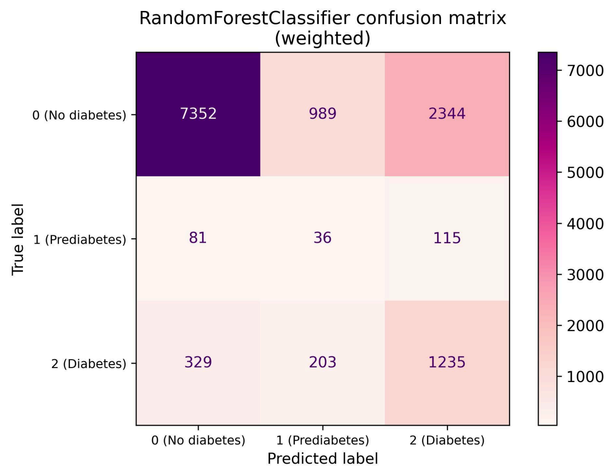
4 Results

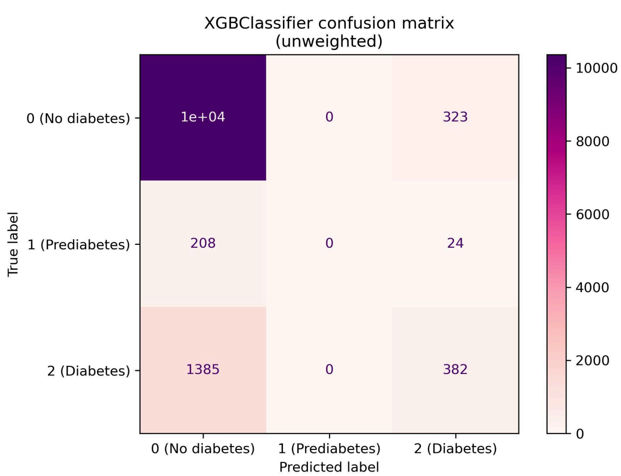
Originally, none of the models had their class_weights parameter set, so over half of them never predicted class 1 (prediabetes) as it was the smallest class and they would be able to minimize overall error by never predicting class 1. Therefore, class-weighted versions (balanced weights) of Random Forest Classifier, XGBClassifier, and LinearSVC were also trained through the pipeline and Figure 5 shows the confusion matrices for both unweighted and weighted versions of these models, along with the matrices for Logistic Regression and K Neighbors Classifier, on the test set for the last random state.

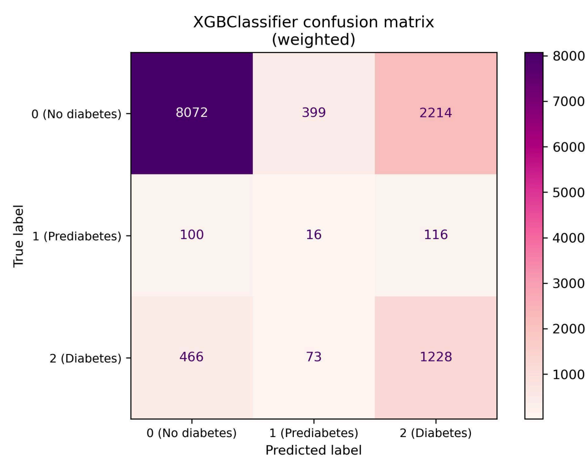

LogisticRegression confusion matrix
(unweighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
1e+04
0
212
218
0
14
1428
0
339
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
10000
8000
6000
4000
2000
0


KNeighborsClassifier confusion matrix
(unweighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
9782
72
831
195
2
35
1253
30
484
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
8000
6000
4000
2000


RandomForestClassifier confusion matrix
(unweighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
1e+04
5
426
198
0
34
1353
3
411
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
10000
8000
6000
4000
2000
0


RandomForestClassifier confusion matrix
(weighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
7352
989
2344
81
36
115
329
203
1235
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
7000
6000
5000
4000
3000
2000
1000


XGBClassifier confusion matrix
(unweighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
1e+04
0
323
208
0
24
1385
0
382
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
10000
8000
6000
4000
2000
0


XGBClassifier confusion matrix
(weighted)
True label
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
8072
399
2214
100
16
116
466
73
1228
0 (No diabetes)
1 (Prediabetes)
2 (Diabetes)
Predicted label
8000
7000
6000
5000
4000
3000
2000
1000

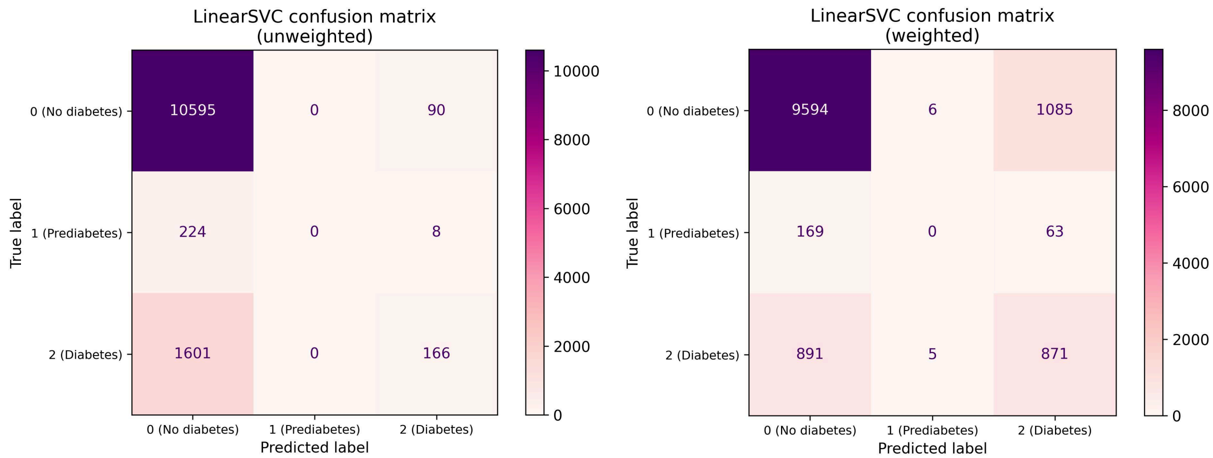


Figure 5. Confusion matrices for the 5 ML Models and their class-weighted versions where applicable

To highlight the need for using f1 macro score as the primary evaluation metric, Figure 6 shows the mean test accuracy for all ML models (unweighted and weighted) as well as the baseline accuracy. All the unweighted models had relatively similar accuracy scores and scored similarly to the baseline accuracy of 0.84. We see that these models are able to achieve a high accuracy even with poor performance on predicting class 1 due to the sheer size of the majority class, which dominates the score. The f1 micro scores and accuracy scores were unsurprisingly equivalent since they do the same calculation.

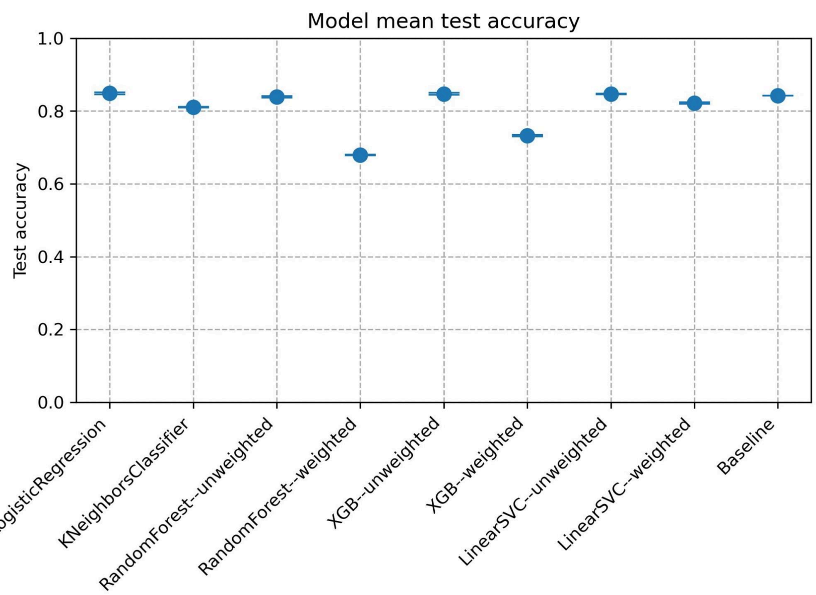


Figure 6. ML models' mean test accuracy vs. baseline accuracy

The f1 macro score is therefore more revealing and is depicted in Figure 7. Weighted LinearSVC had the best mean f1 macro score of 0.45 and standard deviation of 0.002, with the score being 50% higher (about 70 LinearSVC standard deviations above) than the baseline f1 macro of 0.30. Thus, while the class-weighted counterparts of the models always had lower accuracy scores than their unweighted versions, their f1 macro scores are significantly higher. Overall, all models had a f1 macro score greater than the baseline f1 macro.

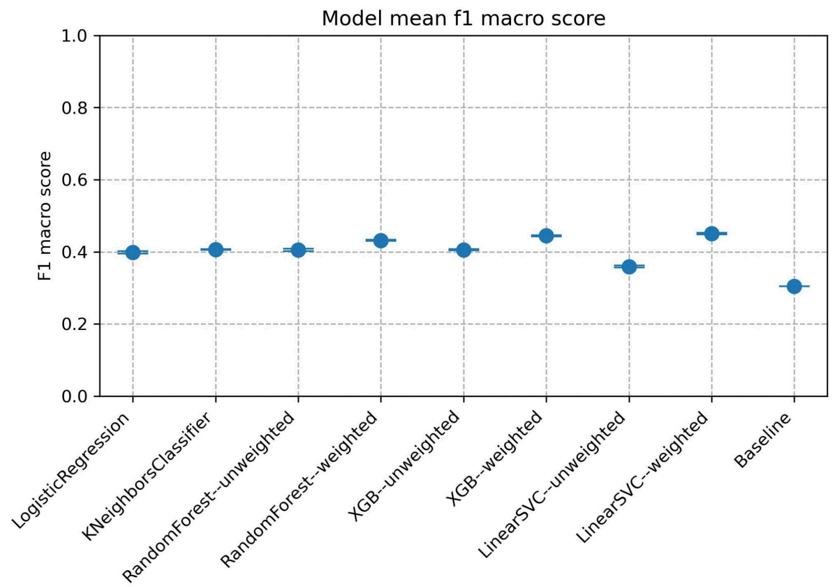


Figure 7. ML models' mean f1 macro score vs. baseline f1 macro

In Figure 8, we see that the weighted LinearSVC had the highest f1 weighted score, with all except weighted Random Forest Classifier scoring higher than the baseline.

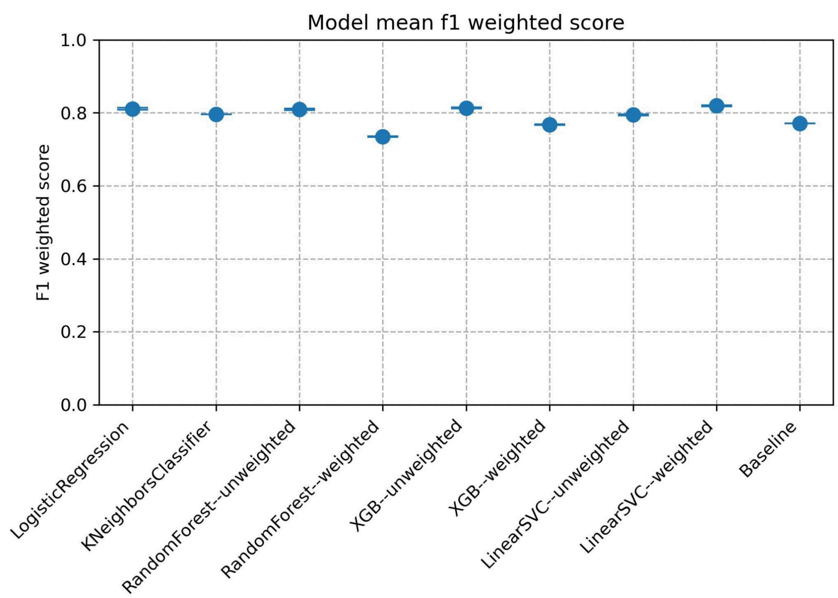


Figure 8. ML models' mean f1 weighted score vs. baseline f1 macro

To understand which features were most important in the best performing model, permutation feature importance, coefficient feature importance, and SHAP global feature importance were calculated. Figure 9 shows the top 10 features when using permutation feature importance for the weighted LinearSVC model, where each feature's values were shuffled one at a time over 10 random states to measure the resulting difference in f1 macro. The top feature in this case is GenHlth, followed by BMI.

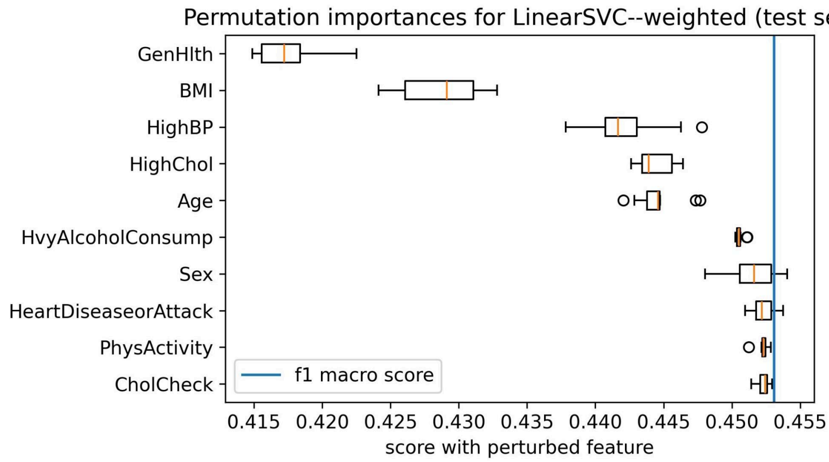


Figure 9. Permutation importances for weighted LinearSVC

Figure 10 shows the top features by coefficients for the weighted LinearSVC model for each class. Figure 11 shows the overall top features by coefficients across all classes. Interestingly, both GenHlth and BMI appear in the top 2 for class 0 (no diabetes) and class 2 (diabetes), but for class 1 (prediabetes), Age is the top feature. To note, GenHlth is ordinally encoded with the value 1 being "Excellent" and 5 being "Poor," explaining the negative coefficient for class 0 (inverse relationship between magnitude and likelihood of prediction). In Figure 11, we see that the overall top features by coefficient are, in order, GenHlth, BMI, and Age.

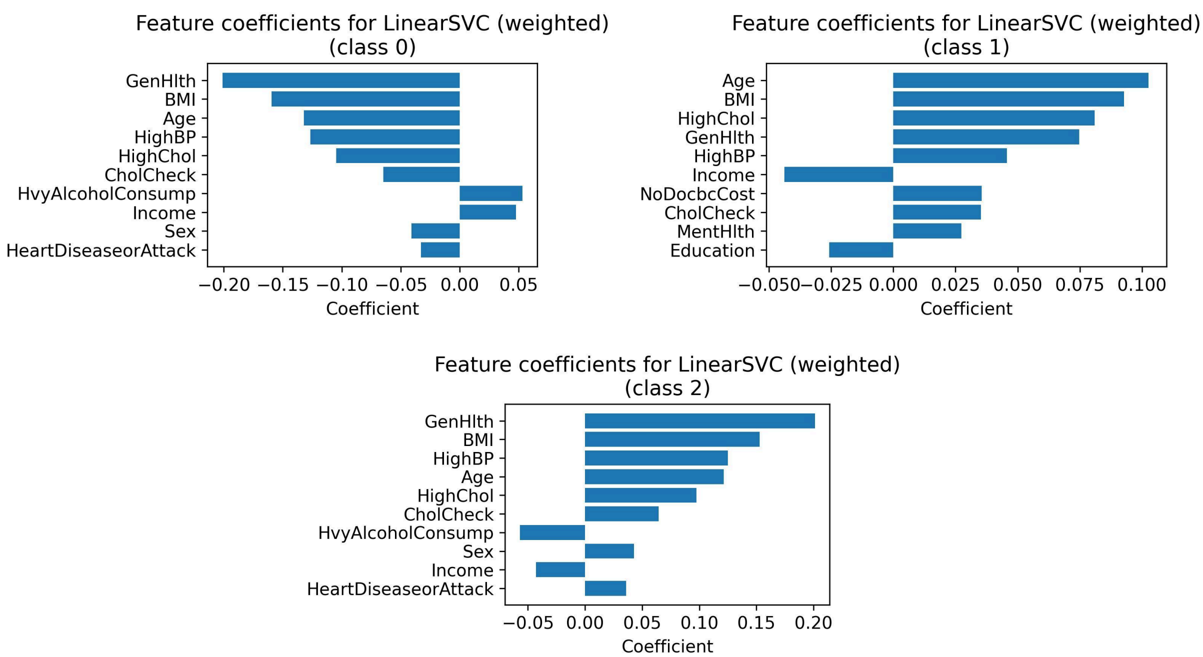


Figure 10. Weighted LinearSVC's feature importance per class by feature coefficients

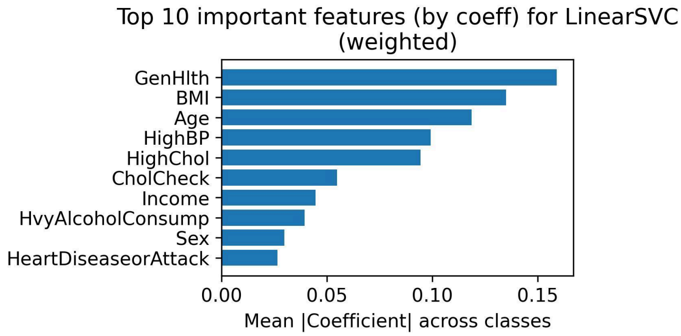


Figure 11. Weighted LinearSVC's overall top 10 features by feature coefficients

Figure 12 shows the SHAP global feature importances, calculated by taking the sum of absolute SHAP value across all samples and classes. Once again, GenHlth, BMI, and Age rank high in importance, and, due to their presence across all metrics, play an important role in prediction for the LinearSVC model. This lines up with medical research on health factors that influence diabetes [5].

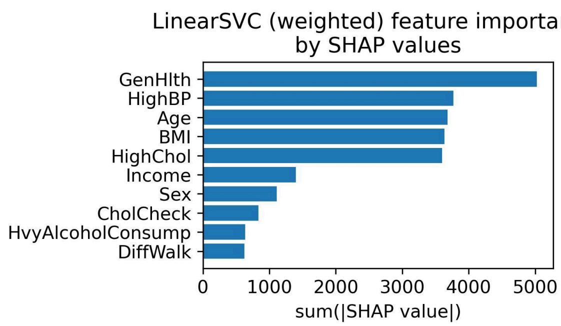


Figure 12. Top 10 features using SHAP global feature importance for weighted LinearSVC

To understand local feature importances, force plots for a single observation in the test set were created and Figure 13 shows how features influence the prediction for each class. For class 0 (no diabetes), features like Age and CholCheck push the prediction toward class 0. For classes 1 and 2, these same features push the prediction downward, revealing how this individual's feature values are most predictive of not having diabetes.

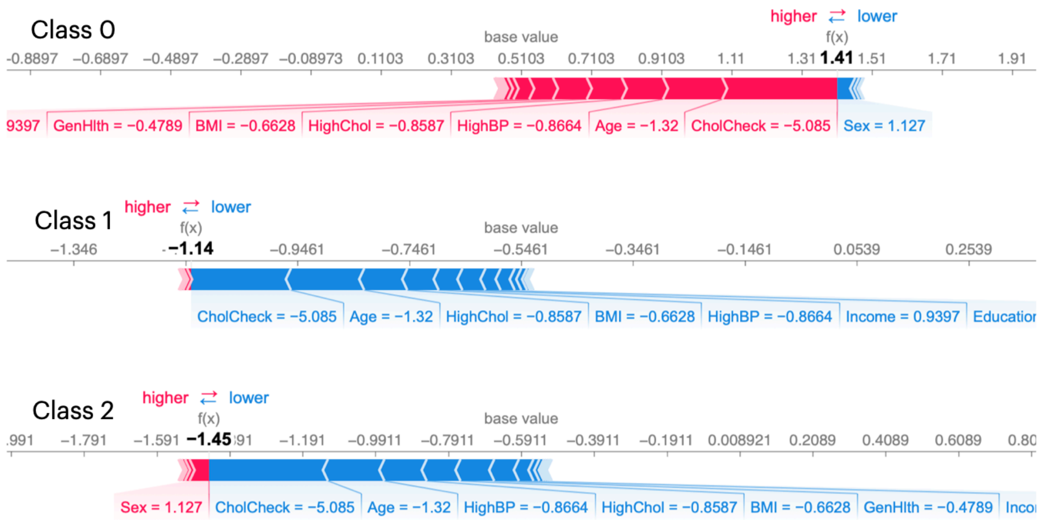


Figure 13. SHAP feature importance by class for single observation (LinearSVC, weighted)

5 Outlook

Table 2 shows the precision, recall, and f1 score for each class. We see that while the f1 macro of the weighted LinearSVC was much higher than the baseline f1 macro score, it is not correctly predicting any points that belong to class 1 and performs only moderately well for class 2. This is a result of the strong class imbalance in the dataset and limits the usefulness of the model in being able to accurately predict whether an individual has no diabetes, prediabetes, or diabetes. Therefore, a future improvement could be to use resampling methods, such as SMOTE, to synthetically add more samples for the minority classes, allowing models to better distinguish between classes. Additionally, for the Logistic Regression model, the class_weights parameter was not able to be used due to it being too computationally expensive, so further exploration can be done here. Lastly, for some of the models, the best values for hyperparameters fall on the borders, so it could be worthwhile to increase the range of values for some parameters.

LinearSVC (weighted) Classification Report

	Precision	Recall	F1 Score	Support
Class 0 (No diabetes)	0.90	0.90	0.90	10,685
Class 1 (Prediabetes)	0.00	0.00	0.00	232
Class 2 (Diabetes)	0.43	0.49	0.46	1,767

Table 2. Classification report for weighted LinearSVC model

6 References

[1]*Diabetes*. (2023, February 17). Cleveland Clinic. Retrieved December 11, 2025, from https://my.clevelandclinic.org/health/diseases/7104-diabetes

[2]*National Diabetes Statistics Report*. (2024, May 15). Centers for Disease Control and Prevention. Retrieved December 11, 2025, from https://www.cdc.gov/diabetes/php/data-research/index.html
[3]Chowdhury, M. M., Ayon, R. S., & Hossain, M. S. (2024). An investigation of machine learning algorithms and data augmentation techniques for diabetes diagnosis using class imbalanced BRFSS dataset. *Healthcare Analytics, 5*, 100297. https://doi.org/10.1016/j.health.2023.100297
[4]Teboul, A. (2022). *Diabetes Health Indicators Dataset* [Data set]. *Kaggle*. Retrieved December 11, 2025, from https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset
[5]*Diabetes Risk Factors*. (2024, May 15). Centers for Disease Control and Prevention. Retrieved December 11, 2025, from https://www.cdc.gov/diabetes/risk-factors/index.html